

Equations

Week 3B

Overview

Quadratic Equations

Key Ideas - Solving Quadratic Equations by Completing the Square

- Solving Quadratic Equations using the Quadratic Formula*
- Equations Reducible to Quadratic Form*
- Solving Equations with Powers*

Key Idea – Solving Quadratic Equations by Completing the Square

The technique known as “completing the square” **can be used to solve all quadratic equations**, whereas factorising is only possible in some cases. Completing the square is also used in many other areas of Mathematics.

Discussion Questions:

The expansion of the perfect square $(x + 5)^2$ is $x^2 + 10x + 25$.

What do you notice about the coefficient of x and the value of the constant term in the expanded form compared with the value of the constant in the factorised form?

factored form
• constant = 5

expanded form
• coefficient of $x = 10$
• constant = 25

- The constant in factored form squared is the constant in expanded form
- The constant in factored form $\times 2$ is the coefficient of x in expanded form

So if we start with $x^2 + 10x$, how do we find the constant term that gives us an expression which is equal to the perfect square $(x + 5)^2$?

$$x^2 + 10x + 5^2 = (x + 5)^2$$

halve the coefficient of x and square it

Steps for solving a quadratic equation by **completing the square**:

Step 1: Make the coefficient of x^2 equal to 1

Step 2: Move the constant term to the right-hand side

Step 3: Halve the coefficient of x and square the result, then add this value to both sides to complete the square

Step 4: Write the quadratic expression as a perfect square.
(the expression is “ x plus half the coefficient of the x term”)

Step 5: Take the square root of both sides and then solve the equation.

Example: Solve $x^2 - 6x - 4 = 0$ by completing the square.

Step 1 Make the coefficient of x^2 equal to 1.

Step 2 Move the constant term (-4) to the RHS.

$$x^2 - 6x = 4$$

Step 3 Halve the coefficient of x and square it $(-3)^2$, then add 9 to both sides to complete the square

$$x^2 - 6x + (-3)^2 = 4 + (-3)^2$$

$$x^2 - 6x + 9 = 13$$

Step 4 Express the LHS as a perfect square.

$$(x - 3)^2 = 13$$

Step 5 Take the square root of both sides and solve.

$$x - 3 = \pm\sqrt{13}$$

$$x = 3 \pm \sqrt{13} \quad (\text{Answer in exact form})$$

$$x \approx 6.606 \text{ or } x \approx -0.606 \quad (\text{Answer correct to 3 decimal point})$$

Exercise:

Solve the following equation by completing the square: $x^2 + 8x + 9 = 0$

Give your answer in exact form and also correct to 2 decimal places.

step 1 → already done $x^2 + 8x + 9 = 0$

step 2 → $x^2 + 8x = -9$

step 3 → $x^2 + 8x + 4^2 = -9 + 4^2$

step 4 → $(x + 4)^2 = 7$

step 5 → $x + 4 = \pm\sqrt{7}$

$$x = -4 \pm \sqrt{7}$$

$$x = -4 - \sqrt{7}, -4 + \sqrt{7}$$
$$= -6.65, -1.35$$

coefficient of $x = 8$
halve it = 4
square it = $(4)^2$
add to both sides

Example: Solve the following equation by completing the square: $2x^2 - 3x - 3 = 0$

Step 1 Make the coefficient of x^2 equal to 1, divide by the coefficient of $x^2 \rightarrow x^2 - \frac{3}{2}x - \frac{3}{2} = 0$

Step 2 Move the constant term $(-\frac{3}{2})$ to the RHS $\rightarrow x^2 - \frac{3}{2}x = \frac{3}{2}$

Step 3 Halve the coefficient of x and square it $\left(\frac{3}{4}\right)^2$, then add 9/16 to both sides to complete the square

$$x^2 - \frac{3}{2}x + \left(\frac{3}{4}\right)^2 = \frac{3}{2} + \left(\frac{3}{4}\right)^2$$

$$x^2 - \frac{3}{2}x + \frac{9}{16} = \frac{3}{2} + \frac{9}{16}$$

Step 4 Express the LHS as a perfect square $\rightarrow \left(x - \frac{3}{4}\right)^2 = \frac{33}{16}$

Step 5 Take the square root of both sides and solve.

$$x - \frac{3}{4} = \pm \sqrt{\frac{33}{16}} \Rightarrow x = \pm \sqrt{\frac{33}{16}} + \frac{3}{4} \Rightarrow x = \frac{3 \pm \sqrt{33}}{4} \quad (\text{exact form})$$

$$x \approx -0.686 \quad \text{or} \quad x \approx 2.186 \quad (3 \text{ decimal places})$$

Exercise:

Solve the following equation by completing the square: $3x^2 - 4x - 9 = 0$.

Give your answer both in exact form and correct to 2 decimal places.

Step 1:

$$x^2 - \frac{4}{3}x - 3 = 0$$

Step 2:

$$x^2 - \frac{4}{3}x = 3$$

Step 3:

$$x^2 - \frac{4}{3}x + \left(-\frac{2}{3}\right)^2 = 3 + \left(-\frac{2}{3}\right)^2$$

Step 4:

$$\left(x - \frac{2}{3}\right)^2 = \frac{31}{9}$$

Step 5:

$$x - \frac{2}{3} = \pm \sqrt{\frac{31}{9}}$$

$$x = \frac{2}{3} \pm \frac{\sqrt{31}}{3}$$

$$x = \frac{2}{3} - \frac{\sqrt{31}}{3}, \frac{2}{3} + \frac{\sqrt{31}}{3}$$
$$= -1.19, 2.52$$

coefficient of $x = -\frac{4}{3}$
halve it $= -\frac{2}{3}$
square it $= \left(-\frac{2}{3}\right)^2$
add it to both sides

Key Idea – Solving Quadratic Equations using the Quadratic Formula

Using the **method of completing the square** we can solve the general quadratic equation $ax^2 + bx + c = 0$ and thus **develop a formula for solving a quadratic equation**.

Step 1 Make the coefficient of x^2 equal to 1: $x^2 + \frac{b}{a}x + \frac{c}{a} = 0$

Step 2 Move the constant term $\frac{c}{a}$ to the RHS: $x^2 + \frac{b}{a}x = -\frac{c}{a}$

Step 3 Halve the coefficient of x and square it, then add it to both sides.

$$x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = -\frac{c}{a} + \left(\frac{b}{2a}\right)^2$$

Step 4 Express the LHS as a perfect square: $\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$

Step 5 Take the square root of both sides, then solve: $x + \frac{b}{2a} = \frac{\pm \sqrt{b^2 - 4ac}}{2a}$

So, the Quadratic Formula becomes:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example:

Use the quadratic formula to find the exact roots of the equation $5x^2 - 3x - 1 = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad \text{where } a = 5, b = -3, c = -1$$

$$x = \frac{-(-3) \pm \sqrt{(-3)^2 - 4 \times 5 \times (-1)}}{2 \times 5}$$

$$= \frac{3 \pm \sqrt{9 + 20}}{10}$$

$$= \frac{3 \pm \sqrt{29}}{10}$$

$$= \frac{3 + \sqrt{29}}{10} \text{ or } = \frac{3 - \sqrt{29}}{10}$$

Exercise: Use the quadratic formula to solve $5x^2 - 9x = -2$

$$5x^2 - 9x + 2 = 0$$

$$a = 5$$

$$b = -9$$

$$c = 2$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-9) \pm \sqrt{(-9)^2 - 4 \times 5 \times 2}}{2 \times 5}$$

$$= \frac{9 \pm \sqrt{41}}{10} \quad (\text{exact form})$$

Exercise: Use the quadratic formula in the equation $m^2 - 5nm + 6n^2 = 0$ to find m in terms of n .

$$a = 1$$

$$b = -5n$$

$$c = 6n^2$$

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-5n) \pm \sqrt{(-5n)^2 - 4 \times 1 \times 6n^2}}{2 \times 1}$$

$$= \frac{5n \pm \sqrt{25n^2 - 24n^2}}{2}$$

$$= \frac{5n \pm \sqrt{n^2}}{2}$$

$$= \frac{5n \pm n}{2}$$

$$= \frac{5n - n}{2}, \frac{5n + n}{2}$$

$$m = 2n, 3n$$

Key Idea – Equations Reducible to Quadratic Form

In this section we are going to look at equations that are **reducible to quadratic form**. Many hard equations can be reduced to quadratic equations by use of a **simple substitution**.

Example: Solve the equation $(x^2 - 5x)^2 - 2(x^2 - 5x) - 24 = 0$

We can see that the variable portion of the first term (i.e. ignore the coefficient) is the same variable portion of the second term.

Note as well that all we really needed to notice here is that the exponent on the first term was twice the exponent on the second term and the third term is a constant, means that this equation is reducible to quadratic in form.

Example continued: $(x^2 - 5x)^2 - 2(x^2 - 5x) - 24 = 0$

Let $u = x^2 - 5x$, substitute into the equation to get

$$u^2 - 2u - 24 = 0$$

$$(u - 6)(u + 4) = 0$$

$$u = 6 \text{ or } u = -4$$

$$u = 6$$

$$x^2 - 5x = 6$$

$$x^2 - 5x - 6 = 0$$

$$(x - 6)(x + 1) = 0$$

$$x = 6 \text{ or } x = -1$$

$$u = -4$$

$$x^2 - 5x = -4$$

$$x^2 - 5x + 4 = 0$$

$$(x - 4)(x - 1) = 0$$

$$x = 4 \text{ or } x = 1$$

Exercise: Solve $x^4 - 17x^2 + 16 = 0$

let $u = x^2$

$$(x^2)^2 - 17(x^2) + 16 = 0$$

$$u^2 - 17u + 16 = 0$$

$$(u-16)(u-1) = 0$$

$$\downarrow$$
$$u-16=0$$

$$u=16$$

$$x^2=16$$

$$x = \pm\sqrt{16}$$

$$= \pm 4$$

$$\swarrow$$
$$u-1=0$$

$$u=1$$

$$x^2=1$$

$$x = \pm\sqrt{1}$$

$$= \pm 1$$

Key Idea – Solving Equations with Powers

Recall: A non-linear equation is different from a linear equation as the **power of the unknown, x , is NOT 1**. We have already looked non-linear equations in the form of quadratic equations. Now we will focus on equations involving indices that do not follow these forms.

To solve a non-linear equation we apply **inverse operations** in the same way as we do to solve a linear equation, remembering to keep the **equation balanced**.

Recall:

- Inverse operation of **taking a square root is squaring**.
- Inverse operation of **squaring is not simply taking the square root as there are two possible solutions: positive or the negative value**. Therefore, when we want to undo the operation of squaring, we must note that the square root could be positive or negative. That is, **we write $\pm\sqrt{}$** . This is the case for any n^{th} root where n is an even integer.
- Inverse operation of **taking a cube root is cubing**.
- Inverse operation of **cubing is taking a cube root**. With no issues regarding positive or negative values.
- In fact, the positive/negative issue **only arises with even powers**, not odd powers. This is the case for any n^{th} root where n is an odd integer.

Definition: If n and m are positive integers, then

$$x^m = a \rightarrow \begin{cases} \text{if } m \text{ is even then the solution is } x = \pm a^{\frac{1}{m}} \text{ (two solutions)} \\ \text{if } m \text{ is odd then the solution is } x = a^{\frac{1}{m}} \text{ (one solution)} \end{cases}$$

$$x^{\frac{m}{n}} = a \rightarrow \begin{cases} \text{if } m \text{ is even then the solution is } x = \pm a^{\frac{n}{m}} \text{ (two solutions)} \\ \text{if } m \text{ is odd then the solution is } x = a^{\frac{n}{m}} \text{ (one solution)} \end{cases}$$

Example: Solve the following equation: $\frac{x^2 - 13}{18} = 2$

$$x^2 - 13 = 36 \quad \text{Multiplying both sides by 18}$$

$$x^2 = 49 \quad \text{Adding 13 to both sides}$$

$$x = \pm\sqrt{49} \quad \text{Taking the square root of both sides}$$

$$x = \pm 7$$

Note: The power in this example is even. Therefore there are two solutions: positive and negative.

Example: Solve the following equation: $\sqrt{4x - 9} = 7$

$$(\sqrt{4x - 9})^2 = 7^2 \quad \text{Squaring both sides}$$

$$4x - 9 = 49$$

$$4x = 58 \quad \text{Adding 9 to both sides}$$

$$x = \frac{58}{4} = 14\frac{1}{2} \quad \text{Dividing both sides by 4 and simplifying}$$

Example: Solve the following equation: $x^3 + 6 = 5$

$$x^3 = -1$$

Subtracting 6 from both sides

$$x = \sqrt[3]{-1}$$

Taking the cube root of both sides

$$x = -1$$

Note: The power is odd, therefore there is only one solution.

Example: Solve the following equation: $a^{\frac{2}{3}} = 4$

$$\left(a^{\frac{2}{3}}\right)^{\frac{3}{2}} = \left(4^{\frac{2}{3}}\right)^{\frac{3}{2}} \quad \text{Raising both sides to the power of } \frac{3}{2}$$

$$a = 4^{\frac{3}{2}}$$

$$= (\sqrt{4})^3$$

$$= (\pm 2)^3$$

$$= \pm 8$$

Note: The power is even, so there are two solutions (positive and negative).

Example: Solve the following equation: $\sqrt[3]{3x} = 4$

$$\sqrt[3]{3x} = 4$$

$$\left(\sqrt[3]{3x}\right)^3 = 4^3$$

Cubing both sides

$$3x = 64$$

$$x = \frac{64}{3} = 21\frac{1}{3}$$

Dividing both sides by 3

Example: Solve the following equation: $125x^{\frac{-3}{2}} = 8$

$$125x^{\frac{-3}{2}} = 8$$

$$x^{\frac{-3}{2}} = \frac{8}{125}$$

Divide by 125

$$\left(x^{\frac{-3}{2}}\right)^{\frac{-2}{3}} = \left(\frac{8}{125}\right)^{\frac{-2}{3}}$$

Raising both sides to the power of $\frac{-2}{3}$

$$x = \left(\frac{8}{125}\right)^{\frac{-2}{3}}$$

$$= \frac{25}{4}$$

Exercise: Solve the following equations:

(a) $\frac{a^{\frac{3}{4}}}{3} - 2 = 7$

(+2)

$\frac{a^{\frac{3}{4}}}{3} = 9$

($\times 3$)

$\frac{3}{4}$ odd $\rightarrow$ no $\pm$

$a^{\frac{3}{4}} = 27$

$a = (27)^{\frac{4}{3}}$

$= 81$

$\rightarrow \text{even} \rightarrow \text{so } \pm$

(b) $x^{\frac{-2}{3}} = \frac{36}{49}$

$$\begin{aligned} x &= \pm \left(\frac{36}{49} \right)^{-\frac{3}{2}} \\ &= \pm \left(\frac{49}{36} \right)^{\frac{3}{2}} \\ &= \pm \frac{343}{216} \end{aligned}$$